

Element B: Documentation and Analysis of Prior Solution Attempts

Power cables often come unplugged accidentally for many reasons, like getting kicked or pulled from their socket unintentionally. This affects the general public. The goal of our project is to prevent cables from becoming unplugged by accidentally getting kicked or pulled on. This will help people prevent loss of computer data, loss of aborted creations when 3D printing, or loss of electricity to refrigerators and freezers.

People with lots of cables need a way to keep them from unintentionally getting unplugged. Many people have lots of cables, but they can easily be unplugged. We need to find a solution to stop cables from being unplugged.

The snug plug is a plastic clip that goes into the vertical ports in an outlet, so when you plug stuff in, it makes it more secure (*Snug Plug - Your Loose Outlet Fix (25/Pack Glow in the Dark) - Amazon. Com, n.d.*). The benefits of this product include simplicity in use and installation. The downsides of this product are that it is not very durable and is relatively expensive, at about \$1 per outlet for a piece of plastic.



Another solution is a 3D printable box to store a power strip and the cables. It also has a lid with a small opening for the cords to stick out. The benefits are that it is aesthetically pleasing, and you can 3D print it if you have a printer (*NEST | Cable Box,*

2026). The downside is that it does not actually keep the cords plugged in, since it is just a box to make the cords look nice. There is no exact value for this product, so if it uses a half roll of filament, it would cost approximately ten dollars.



The last product is a double-sided adhesive that goes on the outlet that sticks to the plug (*Lenink Loose Outlet Plug Fix, 12 Pack Plastic Snug Socket Secured, Outlet Tightener Insert for Most Loose Outlets, with Back Adhesive*, n.d.). The pros are that it is easy to install and is not flammable, and it also keeps your cables organized. The downsides are that it does not always work well, and some plugs do not fit in it. This product costs 16 dollars.



Currently, our student stakeholder, Z, has to crawl under their desk uncomfortably to plug the cable back in. This takes lots of time and is uncomfortable for the stakeholder.

References

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